

5 The photograph shows a fruit fly.



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(a) The gene for eye colour in this species of fruit fly has multiple alleles.

Homozygous fruit flies have two identical alleles for eye colour.

The table gives some information about these alleles.

Allele	Eye colour of homozygous fruit fly	Depth of colour
+	red	darkest colour ↑ lightest colour
w^{ch}	cherry	
w^b	blood	
w^e	eosin	
w^{apr}	apricot	
w^{iv}	ivory	
w^{cr}	cream	
w	white	



- (i) These alleles are found at the same locus.

State what is meant by the term **locus**.

(1)

- (ii) Explain how these alleles can produce fruit flies with different coloured eyes.

(2)

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- (b) The development of an embryo into either a female or a male fruit fly is determined by two different proteins.

The female specific Dsx-F protein leads to the development of a female fly and the male specific Dsx-M protein results in a male fly.

Both proteins are the product of the same gene.

Explain how this gene can give rise to either a female or a male fly.

(4)

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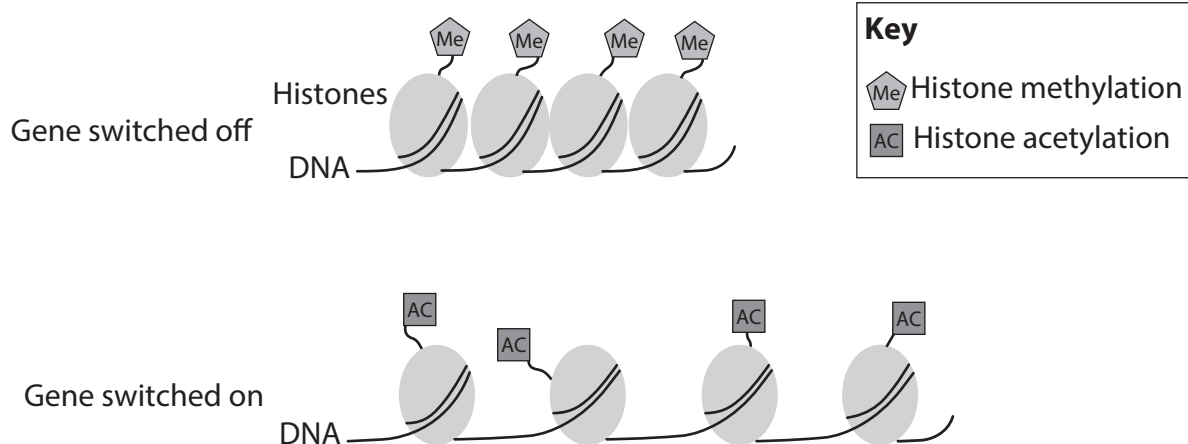
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(c) Histone modification can affect gene expression, as shown in the diagram.



Explain how histone modification can affect gene expression.
Use the diagram to support your answer.

(5)

(Total for Question 5 = 12 marks)

